

ACTIVE GALAXIES

material blasted from the nucleus expands into a lobe as it is slowed by the intergalactic medium

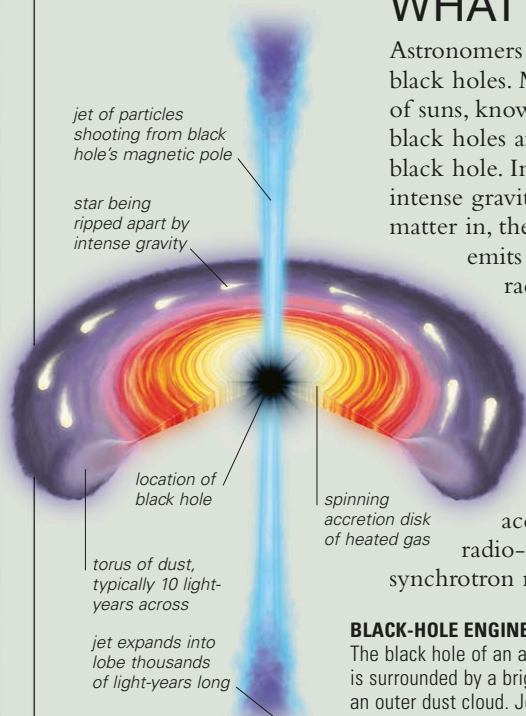
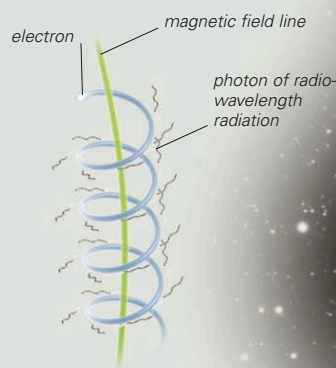
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MANY GALAXIES ACROSS THE UNIVERSE show surprising features that mark them as out of the ordinary. Although there are several types of these strange galaxies, their unusual behavior can always be traced back to powerful activity in their nucleus—it seems that there is an underlying similarity between them, and for this reason, they are often studied together under the term “active galaxies.”

WHAT ARE ACTIVE GALAXIES?

Astronomers think that the features of active galaxies are linked to their central giant black holes. Most, if not all, galaxies have black holes with the mass of many millions of suns, known as supermassive black holes, at their nuclei (see p.305), but most such black holes are dormant—all material in these galaxies is in a stable orbit around the black hole. In active galaxies, matter is still falling inward, and as it falls, it is heated by intense gravity, generating a brilliant blast of radiation. As the black hole “engine” pulls matter in, the superheated material forms a spiraling accretion disk. The hot disk

emits X-rays and other fierce, high-energy radiation. Around the outer edge of the disk, a dense torus (doughnut shape) of dust and gas forms. The intense magnetic field surrounding the black hole also catches some of the infalling material, firing it out as two narrow beams at the poles, at right angles to the plane of the accretion disk. These jets shine with radio-wavelength radiation due to the synchrotron mechanism (right).



BLACK-HOLE ENGINE

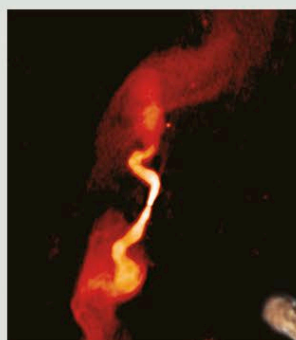
The black hole of an active galactic nucleus is surrounded by a bright accretion disk and an outer dust cloud. Jets of material flow outward from the black hole's poles.

SYNCHROTRON RADIATION

As electrons from the black-hole jets move through the black hole's magnetic field, they are forced into spiral paths, releasing synchrotron radiation—a type of EM radiation that is most intense at long radio wavelengths.

ACTIVE TYPES

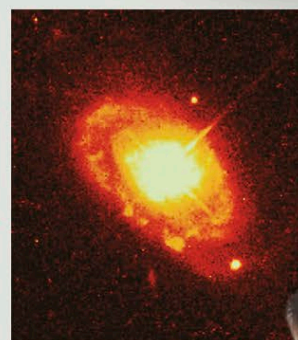
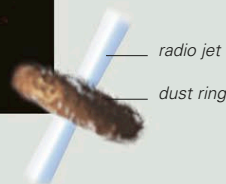
Astronomers distinguish between four major types of active galaxies. Each displays its own set of active features, and in each case, these features are evidence of the violent activity at the nucleus. Radio galaxies are the most intense natural sources of radio waves in the sky. The emissions typically come from two huge lobes on either side of an apparently innocuous parent galaxy (and often linked to it by narrow jets). Seyfert galaxies are relatively normal spirals with a compact, luminous nucleus that may vary in brightness over just a few days. Quasars appear as starlike points of light that show similar but more extreme variability. Red-shifted lines in their spectra reveal that they are extremely distant galaxies—powerful modern telescopes can resolve them as galaxies with incredibly brilliant cores. They are more powerful and more distant cousins of the Seyfert galaxies. Finally, blazars (also known as BL Lacertae objects) are starlike variable points similar to quasars but with no significant lines in their spectra. The standard model of the black-hole engine (above) can explain the major features of each type—how the galaxy appears depends on the intensity of its activity and the angle at which we see it.



RADIO SOURCE 3C31
(RADIO GALAXY NGC 383)

RADIO GALAXY

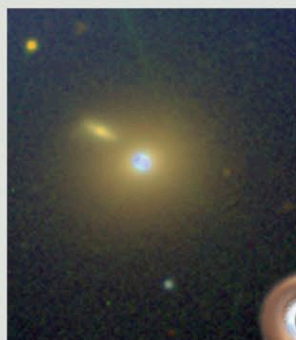
In a radio galaxy such as NGC 383, the central region of the nucleus is hidden by the edge-on dust ring, and observers on Earth see only the radio jets and lobes.



QUASAR PG 0052+251

QUASAR

In quasars, Earth-bound observers can see over the dust ring, and brilliant light from the nucleus and disk drowns out the light of the surrounding galaxy.



BLAZAR MARKARIAN 421

BLAZAR

Blazars are active galaxies aligned so that observers on Earth look straight down the black-hole jet onto the nucleus. The galaxy is usually hidden by the brilliant light except in rare cases, such as Markarian 421.



SEYFERT GALAXY M106

SEYFERT GALAXY

In Seyfert galaxies such as M106, the nucleus and accretion disk are exposed to our view, as in a quasar, but the activity is weak.